

Ring Modulator

The ring modulator originated as a circuit in carrier-frequency technology but soon found diverse applications in electronic music. This circuit is a valuable addition for any adventurous FORMANT player.

Initially, ring modulators were constructed using bifilar-wound coils. However, "double-balanced mixer" ICs have been available on the market for some time, making the construction of ring-modulating circuits a straightforward task (see references). The circuit described here is built around the LM 1496N (or MC 1496P) IC and is suitable for use in the FORMANT system. The choice of the LM 1496N IC is justified by the fact that it is also used in commercial studio synthesizers.

The ring modulator is an example of a four-quadrant multiplier. Such a circuit performs exactly what its name implies: the multiplication of two input voltages. The term "four-quadrant" means that the multiplication remains mathematically correct even when the input voltages take on both positive and negative values. Consequently, the output voltage—following the rules that "minus times minus is plus" and "minus times plus is minus"—can assume both positive and negative values. The oscilloscope photos in Figures 1 and 2, taken using the Elektor ring modulator, provide an idea of what such a circuit can do.

Why, then, is the ring modulator so interesting for use in electronic music? The mathematical relationship for the product of two sine functions is:

$$\sin \alpha * \sin \beta = \frac{1}{2} \cos(\alpha - \beta) - \frac{1}{2} \cos(\alpha + \beta)$$

This equation reflects an interesting phenomenon: multiplying two sinusoidal signals together produces two new sinusoidal signals with frequencies derived from the sum and difference of the two original signals (the "cos," or cosine, is also a sinusoidal signal). Or, expressed in musical terminology: two sine tones multiplied together yield two new sine tones with pitches corresponding to the sum and difference of the original pitches. It should be noted that this applies only to sine tones, not to signals with other wave-forms. However, any arbitrary wave-form can be represented as a combination of sine signals. If, for example, one imagines a combination of two sine signals being multiplied by another sine signal, then each of the two tones in the combination will make its own contribution to the overall signal.

A Different Kind of Harmony

The ring modulator is interesting primarily because it alters the harmonic relationship between tones. We can best understand this through a numerical example. Let us imagine feeding two sine waves—with frequencies of 2.5 kHz and 4.5 kHz, respectively—into one input of a ring modulator. These tones stand in a relationship approximating a so-called diminished seventh, where the frequencies relate as 5:9 (the tones chosen are very high, but this is merely an

example). To the other input of the ring modulator, we feed a sine wave with a frequency of 500 Hz. What happens now at the ring modulator's output? The 2.5 kHz tone, multiplied by the 500 Hz tone, yields—according to the relationship mentioned above—two new tones at 2 kHz and 3 kHz. Similarly, multiplying the 4.5 kHz tone by the 500 Hz tone creates new tones at 4 kHz and 5 kHz. Thus, four tones with frequencies of 2, 3, 4, and 5 kHz are available at the ring modulator's output. Together, these tones form a major chord. In short: the musical relationship of the diminished seventh is transformed into a different musical form—that of a major chord.

However, this description presents a somewhat idealized version of the general situation. In fact, it is relatively rare for musically usable tones at the ring modulator's input to yield musically usable sounds at its output.

The example just described represents an exceptional case. More often, the tones appearing at the ring modulator's output produce a combination of sounds that—to put it mildly—is quite unusual. This becomes even more apparent when we work with signals other than the sine waves used in the example. Periodic signals can be viewed as a combination of a sinusoidal fundamental tone (matching the signal's frequency) and a series of its harmonics. Harmonics are frequencies that are integer multiples of the fundamental frequency.

For example, a sawtooth signal at 1 kHz is composed of sine waves at 1, 2, 3 kHz, etc. If we feed such a sawtooth signal into one input of a ring modulator and a sine wave of, say, 300 Hz into the other, then each harmonic of the sawtooth is multiplied by the 300 Hz sine wave. The resulting output consists of tones with frequencies of 0.7 kHz, 1.3 kHz, 1.7 kHz, 2.3 kHz, 2.7 kHz, 3.3 kHz, etc. This means that the two tones—the 1 kHz sawtooth and the 300 Hz sine wave—produce, via the ring modulator, a whole series of tones that no longer bear a simple relationship to one another. In music, such a combination of tones is called a cluster (or group, swarm)—a simultaneous sounding of a fundamentally different nature from, for instance, a chord.

Even more exotic sounds

A cluster from the previous example becomes even denser if, for the 300 Hz tone, we use a sawtooth wave instead of a sine wave. The resulting variety of tones is almost unimaginable, since—even at the lower end of the frequency spectrum—tones with the following frequencies are produced: 100, 300, 400, 500, 600, 700, 800, 900, 1100, 1200, 1300, 1400, 1500, 1600, and 1700 Hz. Each of these tones has a characteristic amplitude, causing certain tones to strongly dominate. This produces the typical ring-modulator sound—a sound that, if it can be compared to anything at all, is most reminiscent of the timbre of gong-like instruments, or the sound created when metal strikes metal (such as a hammer hitting an anvil).

The potential of a ring modulator is best realized when the frequency of one of the two input signals varies (e.g., is modulated). This produces sounds that undergo dramatic changes in both pitch (as discussed later) and timbre, capable of taking on forms ranging from pure harmonies to piercing dissonances. Highly interesting sounds emerge when the output of a ring

modulator is combined with a noise signal, when the ring modulator is used in conjunction with various filters, or when multiple ring modulators are combined. The composer Stockhausen, for instance, wrote a piece for Hammond organ, chamber choir, and four ring modulators.

Frequency Doubler

The ring modulator can also be used for more "conventional" musical applications. For instance, it can serve as a frequency doubler or octave shifter (since doubling the frequency raises the pitch by one octave). This capability arises when the same signal is fed into both inputs of the ring modulator. The equation demonstrates that, in this case, the difference tone has a frequency of zero Hertz (meaning it does not manifest as an audible tone), while the sum tone has twice the frequency of the shared input signal. When the frequency doubler is used with non-sinusoidal or polyphonic signals, significant intermodulation occurs between the overtones—an effect that, incidentally, offers very interesting musical possibilities. Further applications arise if one of the two input signals has been processed—for example, with an echo effect. Finally, the ring modulator can also be used as a voltage-controlled amplifier (VCA) by connecting the control voltage to one input while using the other input for the signal to be modulated.

A Demanding Instrument

The remarks made here provide essentially just an overview of the musical applications of a ring modulator. All in all, it becomes clear that the ring modulator is a device intended specifically for the experimental musician—one more interested in exploring new sonic textures than in creating conventional harmonies. It is a “demanding” instrument that only advanced musicians can fully exploit to its limits, yet one that is also easily misused. In this respect, the ring modulator could be described as the “violin of electronic music.”

Thus, the ring modulator is part of the standard arsenal of most (reasonably substantial) music synthesizers. Moreover, it is used as a standalone unit by guitarists and keyboard players, as well as by other instrumentalists.

The Ring Modulator Is Not a Ring Modulator

Having concluded that necessarily somewhat digressive explanation and turning now to the more familiar technical aspects, we must note that the name "ring modulator" is actually a misnomer. The term "ring modulator" refers to a specific circuit that does indeed function as a four-quadrant multiplier (at least for AC signals) and was certainly used in the early days of electronic music. Since then, however, electronics technology has produced far superior circuits for this purpose. Nevertheless, the name "ring modulator" has persisted in common usage—after all, most musicians are not interested in the contents of the "black box," but rather solely in what it actually does. A more accurate name for the multiplier used in most cases is

"double-balanced mixer." This is a rather complex circuit consisting of a combination of voltage-controlled current sources. Fortunately, such double-balanced mixers are available as integrated circuits (ICs), so we no longer have to struggle with the process of multiplication—or "ring modulation." To design a ring modulator circuit suitable for musical applications, however, a few additional components are required.

Overview

Figure 3 shows the block diagram of the Elektor ring modulator. The most striking feature is that the ring modulator (indicated by a multiplication symbol) is equipped with an "extra" component: an envelope detector. The reason for adding this otherwise independent circuit was to make efficient use of the available components and PCB space.

The envelope detector outputs the envelope of the input signal. Its input is sensitive enough to process a microphone signal. When used with a synthesizer—such as the FORMANT—this setup allows the envelope of one signal to be applied to another arbitrary signal. At the same time, the envelope detector's input serves (via a summing circuit) as a sensitive input for the ring modulator itself, enabling ring modulation of a signal captured by a microphone (or, for instance, an electric guitar signal). The envelope detector consists of a preamplifier stage and a "block" where the amplified signal is rectified and smoothed, producing a voltage level that corresponds to the input signal's envelope. This process is analogous to standard AM demodulation.

The rest of the block diagram is largely self-explanatory. Of the two mixer inputs, A and B, input B is "split" into a standard input and a high-sensitivity input. Mixer input A is available only as a standard input. The standard inputs are designed for signals up to approx. 1.5 Vpp, such as those found in the FORMANT synthesizer. For the high-sensitivity input, a signal of just 10 mVpp is sufficient for full-scale operation. The sensitivity is therefore high enough for most microphone and guitar pickups. Let us now turn to the detailed circuit shown in Figure 4. The key component is IC1, the double-balanced mixer where the multiplication of the input signals takes place. The IC used here is the LM 1496N. A number of resistors are required to configure the IC for the desired operation. First, the input signals must be kept within safe limits; otherwise, the suppression of the input signals—which should not appear at the output—would be significantly compromised. Therefore, the input signals are attenuated to lower levels (maximum approx. 150 mVpp) using voltage dividers R1/R3 and R8/R11. At this level, the input signals are suppressed by more than 50 dB relative to the output signal. This suppression can be optimized using potentiometers P2 and P3. R6 and R7 form a voltage divider that establishes the required DC voltage level for pins 8 and 10 of the modulator IC. The remaining resistors set the rest of the DC operating points for the LM 1496N. As previously mentioned, the output voltage is matched to the FORMANT standard level of 1.5 Vpp. A3 acts as a buffer.

The envelope detector is constructed using op-amps A1, A2, and A4. A1 is an amplifier circuit that brings the input signal to the desired level. Sensitivity is adjustable via potentiometer P1. Following the amplifier stage, excessively high voltages are limited by D1 and D2. Resistor R9 decouples this signal from the signal at the standard B-input, thereby providing a choice

between the standard and sensitive inputs. Op-amp A2, together with diode D3 and capacitor C6, forms a peak rectifier. Combined with a subsequent low-pass filter having a cutoff frequency of 10 Hz, this produces a signal that precisely tracks the envelope of the input signal. Finally, op-amp A4 determines the level of the envelope. The output voltage of the envelope detector ranges from 0 to 10 V.

Assembly and Alignment

The circuit can be assembled on the PCB shown in Figure 5. Several equivalents exist for the modulator IC. Motorola supplies an IC designated MC 1496P, which is identical to the previously mentioned type and is therefore suitable. Signetics offers the S5596 IC; however, this replacement type is unsuitable for the PCB due to its different pinout. It is best, therefore, to use the LM 1496N from National or the MC 1496P from Motorola.

The circuit operates on supply voltages of +15V and -15V and draws only a few tens of milliamperes. Calibration is simple:

Apply an input signal to input A and adjust potentiometer P3 until the input signal is as inaudible as possible at the ring modulator's output. Then, repeat the process using an input signal at input B and potentiometer P2. After repeating this calibration, the circuit is ready for use, and you can embark on an exciting musical journey of discovery...

If you choose the FORMANT as the platform for this journey of discovery, a front panel in the FORMANT design and a Eurocard-format circuit board are appropriate.

The Front Panel

The proposed front panel design (Figure 6) corresponds in its dimensions to a "small" FORMANT front panel and therefore fits into most FORMANT enclosures. The "Level" potentiometer on the front panel corresponds to P1 in the schematic and is used to adjust the input sensitivity for an external signal. A 6.35 mm jack socket is the most practical choice for ES. The remaining inputs and outputs should be fitted with the 3.5 mm jack sockets standard to the FORMANT system. Arrows schematically indicate the signal paths, allowing even users unfamiliar with the technical details to quickly find their way around.

The Circuit Board

To facilitate installation in 19-inch rack-mount enclosures, the circuit board layout shown in Figure 7 is designed to the Eurocard format. The mounting holes for the front-panel switches and the 31-pin connectors are identical to those used on other FORMANT front panels.

The component layout corresponds to that shown in Figure 4. Interference-suppression electrolytic capacitors can also be installed in the power supply lines.

Assembly Instructions

PCB assembly and alignment present no particular difficulties. In practice, homemade front-panel brackets made of sheet aluminum have proven effective. These are formed from 1.5–2 mm thick sheet metal, as shown in Figure 8. The mounting holes should have a diameter of 3.2 mm. The brackets (two per module) are best attached to the rear of the front panel using a two-component adhesive (e.g., UHU-endfest 300). These brackets can be used for all FORMANT modules and expansion circuits.

Applications

The primary application of the ring modulator is likely the "alteration" or transformation of instrumental and vocal sounds. A few examples illustrate the basic operating principle:

An output signal is generated only when signals are present at both inputs (carrier and modulator) simultaneously. If either signal A or B is missing, no signal appears at the output. The output frequency consists of the sum and the difference of the input frequencies, with neither of the input frequencies being audible on its own (see Table 1). If signals of the same frequency are applied to both inputs, the output signal has double the frequency. When using non-sinusoidal signals, interesting-sounding intermodulation effects also occur.

If the two input signals are tuned an octave apart (frequency ratio 2:1), the output yields a frequency ratio of 3:1 (a duodecime, i.e., an octave plus a fifth). If the ring modulator output is then mixed with the input signals (e.g., using the Universal Mixer, Chapter 5), sounds enriched with additional harmonics are created. Specifically, if the input signals have a frequency ratio of 3:1, harmonics 1, 2, 3, and 4 are produced in this way (Figure 9). In general, new harmonic sounds are created when the frequencies of the input signals stand in an integer ratio to one another.

If rectangular waves are used instead of sine waves—and filtered (using VCFs) either before or after the ring modulator—the result is a sonic "raw material" ideally suited for recreating rich, overtone-laden string, accordion, and bagpipe sounds (Figure 10). A musical scale can also be transposed using a ring modulator; depending on the carrier frequency, the scale may contract or expand.

By suitably selecting the input frequencies, it is possible to generate vowel-like sounds, shortwave-like noises, and even gong or bell-like sounds (see Fig. 11)—all characterized by a predominance of inharmonic overtones. The dissonance becomes more pronounced as the frequency increases and the signal overtones become more densely packed (e.g., sawtooth waves).

Since the Elektor ring modulator features its own preamplifier, the ES input can easily be used to experiment with the sonic transformation of natural instrumental sounds (see Figure 12). In

addition to electro-acoustic and fully electronic instruments, wind instruments (via microphone) are particularly suitable for this purpose. However, depending on the application, additional filtering may be required to suppress generally unwanted frequencies such as $3f_1 - f_2$, $3f_1 + f_2$, $3f_2 - f_1$, $3f_2 + f_1$, etc. Here, too, a portion of the original signals should be mixed in with the ring-modulated signal. Tape and record recordings can also be subjected to sonic transformation via the ES input.

Speech-altering effects—for which the Elektor ring modulator, in combination with the FORMANT, is ideally suited—deserve a chapter of their own. Modulating a microphone-fed voice signal with a low-frequency square wave (see Figure 13) produces a robotic or computer-like voice of the kind familiar from science-fiction films. Additional modulation (e.g., PWM) allows the signal to be distorted further until speech intelligibility is completely lost.

It is also possible to use the ring modulator as a type of "VCA" (two-quadrant multiplier) by feeding the audio signal to be modulated into one input while connecting a control voltage to the other. The potential applications of the envelope follower—also built into the Elektor ring modulator—are described in greater detail in Chapter 5 (Envelope Follower).

Auditory impressions can be described only inadequately in words. In conclusion, therefore, attention is drawn to works by well-known composers who have utilized ring modulators—works that are also available on record:

- W. Kaegi: "Thai Clarinet"
- K. Stockhausen: "Kontakte"

Using multiple ring modulators also increases the possibilities for (timbral) transformation. However, going into further detail would exceed the scope of this description.

The examples provided—some illustrated by "program sheets"—are intended to facilitate the initial steps into experimentation; for the ring modulator, despite (or perhaps precisely because of) its diverse capabilities for actively shaping the sound, can easily be misused and lead to a "musical dead end."

Here are some technical specifications for the ring modulator:

Frequency range: 20... 20,000 Hz

Crosstalk attenuation (between inputs): 60 dB.

Input impedance: 50 k Ω

Output impedance: 500 Ω

Literature:

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